

# Patient Feedback Intelligence Platform

Complete PM Strategy Pack

**Version**

1.0 — Discovery Complete

**Status**

**Build Decision: GO**

**Date**

June 2026

**Artifacts included in this document**

- Market & Competitor Analysis
- VMOST (Vision, Mission, Objectives, Strategy, Tactics)
- Business Strategy Analysis
- OKRs — Year 1
- Product Roadmap (Now / Next / Later)
- Target Users — Primary & Secondary
- Jobs-to-be-Done Canvases (3 personas)
- MVP PRD — Risk Alert Module
- Discovery Research Plan
- Discovery Findings Report — All 4 Gates

# 1. Market & Competitor Analysis

## 1.1 Market Moment

The 10 Year Health Plan (published July 2025) creates a step-change in the materiality of patient feedback. For the first time, trust income is directly tied to patient ratings through three mechanisms:

- Clinical team payment tied to feedback scores
- Patient power payments — patients influence whether providers get paid in full
- Public league tables including patient ratings, updated quarterly from summer 2025

Trusts that cannot analyse feedback at scale face direct financial and reputational exposure. The Medium Term Planning Framework (October 2025) already mandates all providers capture near-real-time feedback on at least 5 wards before patient discharge — this is a current compliance requirement, not a future risk.

| Addressable Market                                           | Annual Opportunity                            | Window                                                       |
|--------------------------------------------------------------|-----------------------------------------------|--------------------------------------------------------------|
| ~311 NHS trusts (acute, mental health, community, ambulance) | £18–40M ARR (core NHS).<br>Secondary: 42 ICBs | 18–24 months before a US competitor makes a serious NHS push |

## 1.2 Competitive Landscape

| Competitor               | AI Themes | Risk Alert | NHS Integrations | Board Output   | Our Advantage                                                           |
|--------------------------|-----------|------------|------------------|----------------|-------------------------------------------------------------------------|
| <b>Qualtrics</b>         | Limited   | X No       | X No             | Template       | Generic; no NHS context, no FFT/PALS ingest, no early-warning logic     |
| <b>Civica / Meridio</b>  | X No      | X No       | ✓ Strong         | X No           | Data capture only; no analytical layer. Potential integration partner.  |
| <b>Cemplicity</b>        | Basic NLP | X No       | Partial          | Charts only    | Reporting-focused, not intelligence-focused. No anomaly alerting.       |
| <b>NRC Health</b>        | ✓ Yes     | Partial    | X No             | ✓ Yes          | US product; regulatory/data residency mismatch. No NHS App integration. |
| <b>In-house Power BI</b> | X No      | X No       | Manual           | Static         | Dominant 'competitor' — manual, slow, sampled. 10YHP makes untenable.   |
| <b>Us</b>                | ✓ Full    | ✓ Core     | ✓ All sources    | ✓ AI-generated | Only product combining all four: ingest, understand, alert, act.        |

### 1.3 Market Tailwinds & Headwinds

| Tailwinds                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Headwinds                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• 10YHP makes feedback financially material — direct income linkage</li><li>• NHS App generating 10× more ratings data than legacy FFT</li><li>• CQC Well-Led inspections increasingly scrutinise feedback use</li><li>• Every major care scandal cited unread patient feedback as warning sign</li><li>• ICB consolidation creates demand for system-level views</li><li>• NHS England actively seeking market input for a 'holistic insight system'</li></ul> | <ul style="list-style-type: none"><li>• NHS procurement cycles are 12–18 months</li><li>• Data governance / IG requirements add complexity to onboarding</li><li>• Existing Civica/Cerner contracts create inertia at data layer</li><li>• Budget pressure — trusts under financial recovery may defer spend</li><li>• 10YHP implementation timeline subject to political risk</li><li>• Post-Synnovis heightened risk appetite at some trusts</li></ul> |

## 2. VMOST — Strategic Intent

### V — VISION

A NHS where every patient's voice is heard in time to prevent harm — not in an inquiry, after the fact.

### M — MISSION

To turn unstructured NHS patient feedback into real-time intelligence that trust boards can act on — making the patient's voice measurable, comparable, and impossible to ignore.

### O — OBJECTIVES

- Reduce median time-from-feedback-to-action at NHS trusts from ~90 days to under 7 days
- Achieve 80% coverage of all structured and free-text feedback sources per trust within 12 months of go-live
- Enable trusts to demonstrate measurable feedback-driven improvements for CQC inspections
- Build a financially sustainable SaaS business with ARR >£2M by end of Year 2

### S — STRATEGY

Land with acute trusts where financial pressure from the 10YHP is most acute. Win on risk-alerting (a capability no incumbent offers) as the initial wedge. Expand to ICB-level system intelligence once trust-level penetration reaches critical mass. Price as an operational necessity, not a discretionary analytics product.

### T — TACTICS

- NHS App + FFT integration as launch connectors (highest volume, lowest friction)
- Risk-alerting dashboard as hero feature in all sales conversations
- Auto-generated CQC evidence pack as immediate time-to-value for quality teams
- G-Cloud / NHS procurement frameworks listing to reduce procurement friction
- 'You Said, We Did' output as trust-facing proof of ROI for renewal conversations
- Reference site programme with 2–3 early trusts willing to co-author case studies

## 3. Business Strategy Analysis

### 3.1 Strategic Positioning

This is a compliance-adjacent, mission-critical SaaS product — not an analytics tool. The framing matters for pricing, sales, and retention. Position alongside patient safety and clinical governance spend. The sales narrative is 'this is how you protect your income and your reputation under the new NHS settlement,' not 'this is a better dashboard.'

### 3.2 Value Framework

| Value Creation                                                                                                                                            | Defensibility                                                                                                                             | Growth Motion                                                                                                                                                      | Go-to-Market                                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Converts a previously unanalysable data asset (free-text feedback) into financial and reputational protection. The value is existential, not incremental. | Moat comes from NHS-specific training data, workflow integrations, and switching costs that grow as historical baseline data accumulates. | Land in a trust with FFT alerting, expand horizontally (complaints, PALS, surveys), then vertically to ICB. Classic land-and-expand in a compliance-driven market. | Direct enterprise sales to Chief Nurse or Chief Quality Officer. Secondary via NHS regional networks and consultancy partners (KPMG, Carnall Farrar). |

### 3.3 Porter's Five Forces

| Force               | Level   | Analysis                                                                                                                       |
|---------------------|---------|--------------------------------------------------------------------------------------------------------------------------------|
| Competitive rivalry | Low–Med | No direct competitor yet. Window of 18–24 months before a US player makes a serious NHS push.                                  |
| New entrants        | Medium  | Low barriers to basic NLP, but high barriers to NHS procurement relationships, IG compliance, and clinical domain expertise.   |
| Supplier power      | Low     | Underlying AI/LLM costs are commoditising. NHS data sources are mandated open through NHS API standards.                       |
| Buyer power         | High    | Long procurement cycles and incumbent loyalty, but the 10YHP changes trust incentive structures dramatically in our favour.    |
| Substitutes         | Medium  | Manual Excel sampling is the primary substitute. Its viability collapses as financial consequences of poor scores materialise. |

## 4. OKRs — Year 1

Four objectives across product, commercial, technical, and feedback loop pillars. Key Results are measurable, time-bound, and validated through the discovery phase.

### O1 — Prove the product works at scale in a real NHS trust

**KR 1:** 2 NHS trusts live on the platform with >80% of feedback sources ingested by end of Q2

**KR 2:** Median time from feedback ingestion to surfaced theme <24 hours (vs status quo of 30–90 days)

**KR 3:** At least 1 confirmed early-warning alert validated by the trust as clinically meaningful by end of Q3

**KR 4:** NPS from quality leads and chief nurses at pilot sites  $\geq 45$

### O2 — Build the commercial foundation for scalable growth

**KR 1:** ARR of £500k by end of Year 1, on a path to £2M by end of Year 2

**KR 2:** At least 1 published case study co-authored with a reference trust, with quantified outcomes

**KR 3:** Listed on G-Cloud and NHS procurement frameworks by end of Q3

**KR 4:** 5 trusts in active procurement (at commercial terms stage) by end of Year 1

### O3 — Establish the technical and data infrastructure for scale

**KR 1:** NHS App, FFT, PALS, and complaints integrations all production-ready by end of Q2

**KR 2:** Platform achieves NHS DSP Toolkit compliance and DTAC assessment by end of Q1

**KR 3:** AI theme classification accuracy  $\geq 91\%$  (validated against manual expert coding on 500-comment sample)

**KR 4:** Platform uptime  $\geq 99.5\%$  on a rolling 90-day basis from Q2 onwards

### O4 — Create clear feedback loops from insight to improvement

**KR 1:** 'You Said, We Did' report used by at least 1 trust to communicate improvement back to patients

**KR 2:** Board-ready pack exported and used in a trust board meeting by at least 2 pilot trusts

**KR 3:** At least 1 trust cites platform outputs as CQC evidence in a Well-Led assessment

## 5. Product Roadmap

Now / Next / Later — phased around the 10YHP implementation timeline and DTAC compliance as a prerequisite to everything.

| Layer             | NOW Q1–Q2 Foundation                                                                                                                | NEXT Q3–Q4 Scale                                                                                                                                      | LATER Year 2+ Expand                                                                                                          |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Ingest</b>     | <ul style="list-style-type: none"> <li>NHS App ratings</li> <li>FFT free text (API + CSV)</li> <li>CSV complaints upload</li> </ul> | <ul style="list-style-type: none"> <li>PALS API</li> <li>Survey platform connectors</li> <li>Datix integration</li> </ul>                             | <ul style="list-style-type: none"> <li>Real-time streaming ingest</li> <li>ICB aggregation layer</li> </ul>                   |
| <b>Understand</b> | <ul style="list-style-type: none"> <li>AI theme classification</li> <li>Sentiment scoring</li> <li>Ward/pathway mapping</li> </ul>  | <ul style="list-style-type: none"> <li>Severity classification</li> <li>Multi-language support</li> <li>Trend detection (QoQ)</li> </ul>              | <ul style="list-style-type: none"> <li>Cross-trust benchmarking</li> <li>Predictive risk modelling</li> </ul>                 |
| <b>Alert</b>      | <ul style="list-style-type: none"> <li>Sentiment deterioration alerts</li> <li>Weekly digest emails</li> </ul>                      | <ul style="list-style-type: none"> <li>Threshold-based real-time alerts</li> <li>Slack/Teams integration</li> <li>Alert audit trail</li> </ul>        | <ul style="list-style-type: none"> <li>Proactive risk scores</li> <li>Serious incident correlation</li> </ul>                 |
| <b>Act</b>        | <ul style="list-style-type: none"> <li>Board-ready pack (PDF)</li> <li>CQC evidence export</li> </ul>                               | <ul style="list-style-type: none"> <li>'You Said, We Did' generator</li> <li>Improvement action tracker</li> <li>API for existing BI tools</li> </ul> | <ul style="list-style-type: none"> <li>Closed-loop improvement tracking</li> <li>Patient-facing outcomes reporting</li> </ul> |
| <b>Platform</b>   | <ul style="list-style-type: none"> <li>DSP Toolkit compliance</li> <li>DTAC assessment</li> <li>Role-based access</li> </ul>        | <ul style="list-style-type: none"> <li>G-Cloud listing</li> <li>SSO / NHS Login</li> <li>Audit logging</li> </ul>                                     | <ul style="list-style-type: none"> <li>Multi-trust tenancy</li> <li>On-premise deployment option</li> </ul>                   |

Key sequencing decision: DSP Toolkit and DTAC compliance are prerequisite to everything — no NHS trust will sign a data processing agreement without them. These are Q1 deliverables, not Q2. Risk-alerting should go live at the same time as the MVP, because it's the feature that justifies the budget in procurement conversations.



## 6. Target Users

### 6.1 Primary Users — Daily / Weekly Platform Interaction

| Chief Nurse / Director of Patient Experience                                                                                                                                                                                                                                                                                                                          | Quality Manager / Patient Experience Lead                                                                                                                                                                                                                                                                                              | Ward Manager / Service Line Lead                                                                                                                                                                                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Strategic owner · Board-facing · Financially exposed under 10YHP</p> <p><b>NEEDS</b></p> <ul style="list-style-type: none"> <li>Know which wards are deteriorating before board meetings</li> <li>Demonstrate feedback-driven improvement to CQC</li> <li>Produce board packs in minutes, not days</li> <li>Early warning of income exposure from 10YHP</li> </ul> | <p>Operational owner · Daily platform user · Process builder</p> <p><b>NEEDS</b></p> <ul style="list-style-type: none"> <li>Analyse 100% of comments — not a sample</li> <li>Theme drilldown to verbatim comments</li> <li>CQC evidence pack in hours, not weeks</li> <li>Closed loop: feedback → action → outcome tracking</li> </ul> | <p>Frontline responder · Alert recipient · Advocate or blocker</p> <p><b>NEEDS</b></p> <ul style="list-style-type: none"> <li>Weekly email digest — no login required</li> <li>Specific, actionable themes not just a score</li> <li>Positive comment highlights for staff morale</li> <li>Simple enough for a 12-hour ward shift</li> </ul> |

### 6.2 Secondary Users — Receive Outputs, Do Not Configure

| Trust Board / NED                                                                                        | CQC Inspectors                                                                                                              | ICB Analytics Teams                                                                                      | Patients (indirect)                                                                                                 |
|----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Receives auto-generated board pack. Needs one page with trend, risk flags, and narrative — not raw data. | Receives evidence pack as part of Well-Led inspection. Validates the trust has a functioning feedback intelligence process. | Year 2+ user. Needs system-level rollup across trusts to spot network-wide themes and outlier providers. | Receives 'You Said, We Did' output — proof that their feedback was heard and acted on. Drives future participation. |

Economic buyer vs end user: Chief Nurse or CFO signs the contract (cares about financial risk and CQC outcomes). Quality Manager and Ward Manager are the daily users (care about workflow and time-saving). If the end user finds it burdensome, the economic buyer hears it at renewal.

## 7. Jobs-to-be-Done Canvases

What each persona is actually trying to accomplish — the job they hire our product to do.

### 7.1 Chief Nurse — JTBD Canvas

I need to know which services are in trouble from a patient experience perspective before I walk into the board meeting — not because the media told me.

— Chief Nurse, large acute trust

| Core Job Statement                                                                                                                                                                                                                                                                                                        | Current Workaround (pain)                                                                                                                                                                                                                                                                                                                                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| When I am accountable for patient experience across a large trust, I want to know which services are deteriorating — and why — before they become a serious incident, so I can protect patients and the trust's income.                                                                                                   | <ul style="list-style-type: none"> <li>Manual quarterly sampling of FFT comments by an administrator</li> <li>Complaints review in monthly committee — weeks after the fact</li> <li>Finding out about ward problems from the press</li> <li>Producing CQC evidence by manually collating spreadsheets over 2–3 weeks</li> </ul>                           |
| Functional Jobs                                                                                                                                                                                                                                                                                                           | Emotional & Social Jobs                                                                                                                                                                                                                                                                                                                                    |
| <ul style="list-style-type: none"> <li>Monitor sentiment trends across all wards in one view</li> <li>Get early warning of deteriorating services before escalation</li> <li>Produce board-ready patient experience report in under an hour</li> <li>Demonstrate feedback-driven improvement to CQC inspectors</li> </ul> | <ul style="list-style-type: none"> <li><b>Confidence:</b> Walk into the board knowing what's happening</li> <li><b>Control:</b> Feel on top of a previously unmanageable data source</li> <li><b>Credibility:</b> Be the executive who catches problems early</li> <li><b>Social:</b> Be seen as proactive, not reactive, on patient experience</li> </ul> |

Switch trigger: (a) near-miss where feedback data was available but not seen in time, (b) upcoming CQC inspection where current evidence is thin, or (c) finance director flagging 10YHP payment exposure.

### 7.2 Quality Manager — JTBD Canvas

I spend two days every quarter manually coding FFT comments into a spreadsheet. I know there must be a better way but nobody has built one that actually works with NHS data.

— Quality Manager, acute trust

| Core Job Statement | Current Workaround (pain) |
|--------------------|---------------------------|
|--------------------|---------------------------|

| <p>When I am responsible for turning patient feedback into actionable intelligence, I want to automatically process all feedback sources into themed, prioritised insight — so I can spend my time acting on problems, not finding them.</p>                                                                                        | <ul style="list-style-type: none"> <li>Exports FFT responses to Excel; manually codes themes</li> <li>Samples 10–20% of comments — the rest go unread</li> <li>Creates CQC evidence pack by copying and pasting into Word — 2–3 weeks of work</li> <li>No systematic way to track whether improvements changed feedback</li> </ul>                              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Gains they want                                                                                                                                                                                                                                                                                                                     | Pains to relieve                                                                                                                                                                                                                                                                                                                                                |
| <ul style="list-style-type: none"> <li>100% of comments analysed, not a sample</li> <li>Drilldown from trend → theme → individual comment</li> <li>Closed loop: feedback → action → outcome tracking</li> <li>CQC evidence pack in hours, not weeks</li> <li>Time to do strategic work instead of manual data processing</li> </ul> | <ul style="list-style-type: none"> <li>2–3 days per quarter manually coding comments</li> <li>Only seeing a sample — knowing important signals are missed</li> <li>No way to prove 'You Said, We Did' actions changed anything</li> <li>Having to beg IT for data extracts that take 2 weeks</li> <li>CQC evidence prep as a multi-week annual panic</li> </ul> |

### 7.3 Ward Manager — JTBD Canvas

Send me an email. One. On Monday morning. Short. Tell me what patients said about my ward last week — the good and the bad. I'll read that. I will not log into another system.

— Ward Manager, A&E

| Core Job Statement                                                                                                                                                                                                                                      | Key Design Constraint                                                                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <p>When I am running a ward and accountable for patient experience on my patch, I want to know what patients are saying before problems escalate — so I can address issues early and give my staff the feedback they deserve.</p>                       | <p>Any tool that adds time to a 12-hour ward shift will be abandoned. Design for the ward manager who has 3 minutes on a Tuesday morning.</p> |
| <p><b>Unexpected insight: Positive feedback is a killer feature</b></p>                                                                                                                                                                                 |                                                                                                                                               |
| <p>'My team never hear the good stuff — only complaints reach us. That alone would change how my staff feel about Monday mornings.' The weekly digest must always include at least one positive verbatim comment. This is structural, not optional.</p> |                                                                                                                                               |

## 8. MVP PRD — Risk Alert Module

Version 0.1 · Status: Discovery validated · Updated with 4 post-discovery pivots

### 8.1 Problem Statement

NHS trusts receive thousands of patient feedback comments per month across FFT, NHS App, PALS, and complaints. No trust has the capacity to read all of it. Emerging patient experience problems go undetected until they escalate to a serious incident, a formal complaint, or a CQC inspection finding. Under the 10YHP and the Medium Term Planning Framework, trusts now face direct income consequences for poor feedback scores and a compliance mandate for near-real-time ward monitoring.

### 8.2 MVP Scope

| ✓ In scope (MVP)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | ✗ Out of scope (MVP)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>NHS App ratings + FFT free-text (CSV + API ingest)</li> <li>AI theme classification on every comment (not sampled)</li> <li>Sentiment scoring per comment, aggregated per ward</li> <li>Trend detection: week-on-week and rolling 4-week sentiment change</li> <li>Alert engine: top-3 deteriorating wards (post-discovery pivot)</li> <li>Weekly digest email to ward managers (no login required)</li> <li>Alert dashboard for Quality Manager — web app</li> <li>Drilldown: alert → theme → verbatim comments</li> <li>Basic board-pack export (PDF)</li> <li>Role-based access: QM + Chief Nurse views</li> </ul> | <ul style="list-style-type: none"> <li>PALS, Datix, complaints integrations (Next phase)</li> <li>Survey platform connectors (Next)</li> <li>'You Said, We Did' generator (Next)</li> <li>Improvement action tracker (Next)</li> <li>ICB / cross-trust aggregation (Later)</li> <li>Real-time streaming ingest (Later)</li> <li>Multi-language support (Later)</li> <li>On-premise deployment (Year 2 — architecture decision needed now)</li> <li>Ward-facing web app (Later — digest is the product for WMs)</li> </ul> |

### 8.3 User Stories (Must Have)

#### US-01 — Must Have

As a Quality Manager, I want to see a dashboard showing the top 3 wards with the biggest sentiment deterioration in the last 4 weeks — not all wards above threshold — so I can prioritise my attention without alert fatigue.

**ACCEPTANCE CRITERIA**

- a) Dashboard shows top 3 wards by sentiment delta (most deteriorated first) by default
- b) Each row shows: ward name, current sentiment score, 4-week change, comment volume
- c) Red/amber/green indicator based on configurable threshold
- d) 'Show all wards' toggle reveals full trust view
- e) Clicking a ward opens the theme drilldown view

**US-02 — Must Have**

As a Quality Manager, I want to drill from an alert into the specific themes driving the change, so I know what to address.

**ACCEPTANCE CRITERIA**

- f) Theme view shows top 5 themes contributing to the change, with volume and delta
- g) Each theme shows representative verbatim comments (min 3, max 10)
- h) Comments sorted by relevance score, with previous week's comparison visible
- i) Classification confidence score shown; QM can correct a misclassification

**US-03 — Must Have**

As a Ward Manager, I want a weekly email digest with last week's patient feedback about my ward — including positive comments — delivered without needing to log in.

**ACCEPTANCE CRITERIA**

- j) Email sent every Monday 08:00 covering previous 7 days
- k) Contains: sentiment direction, top 2 positive themes, top 2 concern themes, 3 verbatims (min 1 positive)
- l) One-tap 'share with team' link opens stripped-down version
- m) Magic link expiry: 7 days minimum
- n) If no comments that week: email still sent — never silently skipped

**US-04 — Must Have**

As a Chief Nurse, I want an immediate alert when any service crosses the red threshold, so I can act before the board meeting.

**ACCEPTANCE CRITERIA**

- o) Alert delivered via email within 24 hours of threshold being crossed
- p) Contains: service name, threshold crossed, top driving theme, direct link to dashboard
- q) Alert suppressed 14 days after first trigger on same ward (override available)
- r) Full alert audit log visible in dashboard

## 8.4 Non-Functional Requirements

| Requirement           | Specification                                                                                                                                      |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Data residency</b> | All patient data processed and stored within UK (NHS Azure UK South / UK West). No data transits outside UK.                                       |
| <b>Compliance</b>     | DTAC assessment complete before any trust data onboarded. UK GDPR Article 9 compliant processing. DPIA signed by trust IG team.                    |
| <b>Availability</b>   | 99.5% uptime on a rolling 90-day basis. Planned maintenance outside 06:00–22:00 Mon–Fri.                                                           |
| <b>Performance</b>    | Dashboard load <2s. Alert emails within 24h. CSV processing <5 min for 5,000 rows.                                                                 |
| <b>AI accuracy</b>    | Theme classification ≥91% overall, ≥88% per theme. Accuracy reported transparently. No comment suppressed — low-confidence flagged, not discarded. |
| <b>Access control</b> | Role-based: System Admin, Quality Manager (trust-wide), Ward Manager (ward-scoped), Chief Nurse (trust-wide read + alerts).                        |
| <b>Accessibility</b>  | WCAG 2.1 AA. Colour is never the sole means of conveying alert status.                                                                             |

## 9. Discovery Research Plan

8-week plan — validate before building. Sequenced to de-risk the biggest unknowns first.

### 9.1 Research Goals (Priority Order)

| Priority               | Goal                                                                                                                      |
|------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>Must validate</b>   | Will NHS IG/CISO sign off on a third-party processor for patient feedback data? This is a build/no-build question.        |
| <b>Must validate</b>   | What does a meaningful, actionable alert look like to a Quality Manager? What triggers would cause them to act vs ignore? |
| <b>Must validate</b>   | Is AI theme classification accurate enough on real NHS free-text without domain-specific fine-tuning? (Technical spike)   |
| <b>Should validate</b> | What does the Ward Manager digest need to contain to drive action rather than deletion?                                   |
| <b>Should validate</b> | What is the right pricing model and budget holder? Is this quality/safety budget or IT/analytics budget?                  |

### 9.2 Interview Plan by Phase

| Phase 1 — Foundational Interviews                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Weeks 1–3 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>4× Chief Nurses, 4× Quality Managers, 3× Ward Managers, 2× IG/CISO leads</b> <ul style="list-style-type: none"> <li>Phase 1 covers the core problem space. Chief Nurse questions focus on feedback lag, 10YHP anxiety, and CQC triggers.</li> <li>Quality Manager questions focus on current process mapping, sampling rates, and alert design preferences.</li> <li>IG/CISO questions focus on data processing approval conditions, DTAC process, and DPIA requirements.</li> </ul> |           |
| Phase 2 — Technical Spike                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Weeks 2–4 |
| <b>Engineering + data science + 1 willing Quality Manager (data access)</b> <ul style="list-style-type: none"> <li>Classify 500 real FFT comments. Validate accuracy against expert coding. Understand confusion matrix by theme.</li> <li>Audit data formats from 3 trusts. Estimate ward naming normalisation effort. Volume and cost benchmarking.</li> <li>Back-test threshold calibration on historical trust data to establish meaningful vs noise signal thresholds.</li> </ul>  |           |

| Phase 3 — Prototype Testing                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Weeks 5–6 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>3× Quality Managers, 4× Ward Managers, 2× Chief Nurses</b> <ul style="list-style-type: none"> <li>• Test alert dashboard: can QMs identify highest-risk ward unprompted in &lt;30 seconds?</li> <li>• Test alert trust: would they act on a red alert? What would make them doubt it?</li> <li>• Test digest engagement: reading time, what they'd forward, what they'd skip.</li> <li>• Test drilldown utility: navigation without help; verbatims reached in &lt;3 clicks.</li> </ul> |           |
| Phase 4 — Pricing & Procurement                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Weeks 6–7 |
| <b>3× Chief Nurses (re-engaged), 2× Finance Directors, 1× NHS procurement lead</b> <ul style="list-style-type: none"> <li>• Van Westendorp pricing: too cheap / acceptable / too expensive thresholds.</li> <li>• Budget location: quality/safety vs IT. Co-signatory requirements above £30k.</li> <li>• Procurement route: G-Cloud direct award vs tender. Threshold and timeline.</li> </ul>                                                                                            |           |

## 9.3 Decision Gates

All four gates must be assessed before writing a line of production code.

| Gate                     | Pass Criterion                                                                                                              |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>IG / CISO Gate</b>    | At least 2 of 4 IG leads indicate they would approve a DTAC-compliant cloud processor. If 0/4: re-scope to on-premise only. |
| <b>AI Accuracy Gate</b>  | Spike achieves ≥85% accuracy on real NHS data without fine-tuning. If not: include fine-tuning in scope or narrow taxonomy. |
| <b>Alert Design Gate</b> | ≥4 of 5 Quality Managers in prototype testing describe a specific action they would take on a red alert.                    |
| <b>Commercial Gate</b>   | At least 1 trust willing to sign a paid pilot at ≥£30k before build. If not: extend discovery, not build.                   |



## 10. Discovery Findings Report

8 weeks · 14 interviews · 9 prototype sessions · 1 technical spike · Sources grounded in NHS evidence, NIHR research, and 10YHP policy documents

### VERDICT: BUILD — with 4 targeted pivots

All four decision gates pass. The problem is real, well-evidenced, and currently unsolved at scale. The 10YHP has materially increased urgency. NHS England is actively seeking market input on a 'sustainable and holistic insight system.' The technical spike confirms AI classification is achievable. One IG pivot required (on-premise option). Proceed to sprint planning with revised assumptions.

### 10.1 Top 5 Findings That Change What We Build

#### F1 IG gate passes — conditionally

3 of 4 IG leads would approve a cloud processor, but only with UK data residency and DPIA completed before any patient data is transferred. One trust requires an on-premise processing option as a hard condition. We need to offer this or lose that trust segment.

#### F2 AI accuracy reaches 89% out-of-the-box — fixable without full fine-tuning

Sentiment accuracy is strong (94%). Theme accuracy drags to 86% on dignity/respect and multi-theme comments. A revised taxonomy splitting 'dignity and respect' into two concrete sub-categories closes the gap. Estimated: 2 days of engineering.

#### F3 Alert fatigue is the #1 risk — from too many true positives, not false ones

Quality Managers didn't reject false positives; they rejected too many true positives. When every ward shows amber, nobody acts. Only the top-3 deteriorating wards should appear in red by default; everything else invisible until drilled into.

#### F4 Ward managers will not log in — ever

All four ward managers confirmed this unprompted. The email digest is the only viable touchpoint. It must always contain at least one positive comment or it will be read as a criticism tool and resented.

#### F5 The real near-term driver is compliance, not payment linkage

The 10YHP patient power payment is in trial phase, not live. However, the Medium Term Planning Framework already mandates near-real-time feedback on 5+ wards before discharge. This is a current compliance requirement. Lead with this in sales conversations.

## 10.2 Technical Spike Results

| Theme                     | Accuracy | Notes                                                                             |
|---------------------------|----------|-----------------------------------------------------------------------------------|
| Communication             | 92%      | Clear language cues. Model distinguishes well between communication and dignity.  |
| Environment & facilities  | 95%      | Most reliably classified. Highly concrete vocabulary (noise, cleanliness, food).  |
| Waiting & delays          | 93%      | Time language is specific and consistent.                                         |
| Clinical care & treatment | 89%      | Main error: confusing clinical skill feedback with staffing levels.               |
| Staffing & attitudes      | 87%      | Overlaps with communication and dignity themes.                                   |
| Dignity & respect         | 76%      | Weakest category. Dignity language is often euphemistic. Requires taxonomy split. |
| Sentiment (overall)       | 94%      | Strong. Positive/negative/neutral classification highly reliable.                 |
| Overall accuracy          | 89%      | Short of 92% target. Taxonomy change closes to 91–93% without fine-tuning.        |

## 10.3 Gate Verdicts

| Gate              | Verdict            | Evidence & Conditions                                                                                                                                                         |
|-------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IG / CISO Gate    | PASS               | 3/4 IG leads approve cloud processing with conditions: UK residency, ISO 27001, Cyber Essentials Plus, DPIA. 1 trust requires on-premise — add as Year 2 architecture option. |
| AI Accuracy Gate  | PASS (conditional) | 89% overall vs 85% gate criterion. Passes gate. 92% PRD target needs taxonomy change (2-day task). Revise target to ≥91%.                                                     |
| Alert Design Gate | PASS               | 3/3 QMs tested said they would act on a red alert — but only on the top-3 design. Original all-wards design failed unanimously. Pivot confirmed.                              |
| Commercial Gate   | PASS               | 1 LOI signed at £45k (North West acute trust). 2 trusts at commercial terms stage. Gate criterion met. Pipeline projects £200k ARR by Year 1 end.                             |

## 10.4 PRD Pivots Required Before Sprint 1

|                                         |                                                                                                                                                                                                                                                             |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Pivot 1 — Alert UX</b>               | Rewrite US-01. Default alert dashboard shows top-3 deteriorating wards only. All wards accessible via 'show all' toggle. Reason: Alert fatigue confirmed as primary adoption risk. Top-3 design passed; all-wards design failed unanimously.                |
| <b>Pivot 2 — Ward Manager interface</b> | Ward-facing web app deprioritised from MVP to Next phase. Email digest elevated to P0. Add 'share with team' one-tap link. Extend magic link expiry to 7 days. Reason: All 4 ward managers confirmed they will not log in.                                  |
| <b>Pivot 3 — AI taxonomy</b>            | Split 'dignity and respect' into: 'Privacy and dignity' and 'Feeling dismissed or not listened to.' Revise accuracy target to $\geq 91\%$ overall. Reason: Dignity/respect accuracy of 76% is an unacceptable outlier on the most sensitive theme category. |
| <b>Pivot 4 — Architecture</b>           | Add on-premise deployment option to architecture design (Year 2 delivery, architecture decision now to avoid rework). Reason: Post-Synnovis risk appetite shift means ~15–20% of large acute trusts may require this.                                       |
| <b>Addition — Chief Nurse digest</b>    | New US-07 (Should have): Chief Nurse weekly briefing email, trust-wide aggregation of digest template. Estimated engineering: 0.5 days. Identified in prototype testing — not in original PRD.                                                              |

## 10.5 Immediate Actions Before Sprint 1

| Action    |                                                                                                                                                                    |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>A1</b> | Rewrite US-01 and US-03 in the PRD to reflect top-3 alert default and email-primary ward manager interface. Circulate to pilot trust for sign-off.                 |
| <b>A2</b> | Begin DTAC and DPIA process for pilot trust immediately. Appoint a named IG lead this week. Target: DPIA first draft to trust IG team within 3 weeks.              |
| <b>A3</b> | Apply taxonomy change (dignity sub-category split) and re-run classification on 500-comment validation set. Confirm $\geq 91\%$ before production code is written. |
| <b>A4</b> | Schedule ward naming mapping workshop with pilot trust Quality Manager before sprint planning.                                                                     |

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|-----------|-------------------------------------------------------------------------------------------------------------------------|
| <b>A5</b> | Begin G-Cloud application process. Takes 4–8 weeks — start now, not after first contract closes.                        |
| <b>A6</b> | Prepare Finance Director ROI pack: staff hours saved + CQC inadequate rating cost avoidance + 10YHP compliance framing. |